

Energy consumption estimates

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I. INTRODUCTION

A. (Proposed) Hardware Parameters:

I make the following modelling assumptions for the IBM processors:

We use the following hardware parameters in the temperature dependent model:

- γ_0 is fit at runtime to the known values of T_1 at the default operational mixing chamber temperature
 - Assume the operational cryostat temperature to be 0.013K
- $\gamma_{\varphi\text{-base}}$ is fit at runtime to the known values of T_2 at the default operational mixing chamber temperature
 - Again, assume the operational cryostat temperature to be 0.013K
- $\frac{\gamma_{MXX}}{\gamma_0}$ is 0.85.
 - This represents the maximum value from the experimental dataset reported in Ref. [?].
- T_{env} as 30mK
 - Effective temperatures as low as 20mK have been observed Ref. [?].
- Δ (superconducting energy gap) of $2.88e^{-23}$
 - This is the known value for Aluminium, which we can assume the electrodes used in IBM transmon qubits are made of

- ω_{ge} is extracted from known configuration data for each backend individually for each qubit
- N_e , the number of effective channels in the Josephson Junction, is estimated at 1.29×10^5 . We calculate this value using:
 - An estimated normal resistance of the Josephson Junction of $5k\Omega$
 - Transparency of the junction in the subgap regime of 5×10^{-5}
- E_c , charging energy of the capacitors in the transmon qubits, of $0.95\mu\text{eV}$
- E_j , josephson energy of the transmon qubits, of $62.1\mu\text{eV}$

Starting with an estimated value of γ_0 : $5e3$, and an estimated standard operational cryostat temperature of 0.013K, we minimise the squared error between the calculated and known, default relaxation time of the qubits. We then do the same for the value of $\gamma_{\varphi\text{-base}}$ keeping γ_0 constant

The final scale factors for the value of T_1 and T_2 for each qubit are stored and applied to the calculated values to maintain granularity

When calculating gate errors, the difference between the known gate error and calculated gate error at the operational temperature is stored and assumed to be representative of the non-thermal error, and thus independent of temperature. It is then added to future calculated gate errors